

RYAN THOMAS

SOFTWARE ENGINEER

Orlando, FL 32804 | 239-595-5555 | rgt219@outlook.com | erregeteygo.com

PROFESSIONAL SUMMARY

Highly skilled **Software Engineer with 6+ years of full-stack engineering experience** architecting, scaling, and deploying high-performance applications across aerospace, defense, and cloud platforms. Proven track record of combining enterprise web development with low-level systems engineering—specializing in **React frontends paired with C# .NET Core and Java REST APIs**, multi-threaded C/C++ simulation, and distributed real-time messaging. Demonstrated success optimizing SQL database query search times by **4+ minutes** at Northrop Grumman, building event-driven streaming pipelines with **Azure Event Hubs (Kafka API) and SignalR WebSockets**, and orchestrating resilient **Generative AI Agents (Gemini LLM)** with automated exponential backoff retries and Sentry telemetry. Proficient in cloud infrastructure, Docker containerization, and **automated CI/CD pipelines via GitHub Actions**.

TECHNICAL SKILLS

Languages & Frameworks:	.NET / .NET Core, C#, C++, Java, React, Python, JavaScript (ES6+), SQL Server, Spring Boot, Qt
Cloud & DevOps:	Azure (Container Apps, Cosmos DB, Event Hubs, Blob Storage), AWS, Docker, GitHub Actions CI/CD, Jenkins, Bamboo
Messaging & AI Tools:	Apache Kafka, RabbitMQ, SignalR WebSockets, Gemini AI SDK, Sentry Telemetry, Web Audio API
OS & Platforms:	Red Hat Enterprise Linux (RHEL 9), VxWorks RTOS, Linux, Git, PowerShell, Visual Studio, TCP/IP

PROFESSIONAL EXPERIENCE

Northrop Grumman — Software Engineer Orlando, FL | Sep 2023 – Present

- **Aircraft Systems Development:** Lead development and design of E-2D Advanced Hawkeye aircraft PCMS and WRMS software systems on Red Hat Enterprise Linux 9 (RHEL 9).
- **Database Optimization:** Reduced query search time of wireless signal data in SQL Server by **4 minutes** by consolidating multi-table join logic.
- **Full-Stack Signal Application (React/Java):** Developed a full-stack solution using React and Java to support the development and real-time signal processing of the WRMS application.
- **Aircraft Display Engineering (Qt/C++):** Built data reduction displays on aircraft hardware architecture using the C++ Qt framework to visualize critical flight diagnostics.
- **Scalable Architecture:** Designed scalable, readable, object-oriented code in .NET Core C#, C++, and Java, incorporating multi-threading and unit tests within each subsystem to ensure software efficiency.
- **Signal Telemetry Streaming:** Incorporated Apache Kafka and RabbitMQ for signal topic partitioning, enabling efficient organization and streaming of telemetry messages.
- **DevOps & CI/CD:** Utilized Git and Bamboo for source control alongside Jenkins for automated testing of microservice build scripts, managing ticket resolutions within an Agile/Atlassian framework.

Coleman Aerospace — Software Engineer Orlando, FL | Feb 2019 – Aug 2023

- **Embedded Simulation:** Built, compiled, and deployed C/C++ simulation code using VxWorks RTOS and Red Hat Linux OS for launch vehicles, satellites, and missile defense power systems.
- **Avionics Documentation & Design:** Developed technical specifications and design documentation associated with avionics test equipment software while enforcing company object-oriented C++ coding standards.
- **Security & Systems Compliance:** Implemented Red Hat Linux updates to align with lab support equipment hardware and updated unclassified/CUI documentation.

FEATURED PROJECT

ErreGeTe YGO — Full-Stack Cloud & AI Platform (erregeteygo.com) Personal Project

- **Full-Stack Architecture & Cloud Infrastructure:** Developed a multi-service web application built with a **React frontend** and an asynchronous **.NET Core C# API backend**, containerized using **Docker** and deployed to **Azure Container Apps** via automated **GitHub Actions CI/CD pipelines**.
- **Event-Driven Pipeline & Real-Time WebSockets:** Engineered an asynchronous data streaming pipeline leveraging **Azure Event Hubs (Apache Kafka API)** to decouple data ingestion from core domain logic, broadcasting live community deck creations and activity feeds via **SignalR WebSockets**.
- **Generative AI Agents & Hand Simulator:** Built interactive AI agents using Gemini LLM to power a **Combo Strategy Playbook & Opening Hand Simulator** (analyzing 5-card draws, combo branches, and Nibiru/chokepoint vulnerabilities) and an automated **Deck Profile Copywriter** generating structured Markdown articles.
- **Resilient AI Proxy & Telemetry:** Architected a resilient .NET C# Gemini proxy featuring **3-stage exponential backoff retries** for rate limits (HTTP 429), safety category overrides for TCG nomenclature, and **Sentry telemetry logging** with custom error tags.
- **Database, CDN & Market Valuation:** Migrated local MongoDB stores to an **Azure Cosmos DB vCore cluster**, configured an **Azure Blob Storage CDN** with image fallback handling, and integrated a live TCGPlayer/Cardmarket API pricing engine.

EDUCATION

Florida State University — Bachelor of Science in Computer Engineering Tallahassee, FL